Big Data Visual Analytics (CS 661)

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Study Materials for Lecture 11

- Visualizing High-Dimensional Data: Advances in the Past Decade; S. Liu et al., TVCG2016
- t-SNE: https://distill.pub/2016/misread-tsne/
- UMAP: https://pair-code.github.io/understanding-umap/

Acknowledgements

- Some of the following slides are adapted from the excellent course materials and tutorials made available by:
  - Prof. Klaus Mueller (State University of New York at Stony Brook)
  - Prof. Tamara Munzner (University of British Columbia)
  - Zaur Fataliyev (Research Scientist – Meta)
  - Andreas Kollegger (neo4j)

Final Project Group Formation

- Form your project team by June 20th and update the google sheet with details of project members
  - https://docs.google.com/spreadsheets/d/17JwSTvZNrZI1waj1ZZDyZaIVwMGkY2fR9SQ_6fzlclo/edit?usp=sharing
  - **Group size: 7/8/9 (8 is preferred)**
- Submit your project proposal by June 23rd in HelloIITK
- Once I approve your proposal, start working on it!
- If I see your work is trivial, insufficient, and does not meet the quality bar during the final presentation, you will not get a good grade for the project
  - So, if you are in doubt, talk to me as you work on the project!

Final Class Project: What is Expected?

- You will build a visual-analytics interface/software to solve a problem from an application domain
- You are expected to have knowledge about your domain of application and the tasks you have picked must reflect that
- Your tasks should be meaningful, and you should be able to explain why you are doing something
- You should be able to tell a coherent story about your data through your visualization interface
  - Random set of plots will not help
  - You need to justify why you picked certain type of plots

Final Class Project: What is Expected?

- You are expected to show meaningful patterns from your data through your interface
- Since this is a visualization focused course, so even if you do a very sophisticated modeling, your visualization interface still should be the main focus for grading
  - How you are presenting your results will matter
- You are expected to write a comprehensive report of your work that describes the details of the methodologies and visualization techniques that you have used
  - You should justify your design choices for your interface
    - Such as: Why a bar chart and not a Pie chart!

Final Class Project: How Will It be Graded?

- Grading will be done on your overall idea + quality of work, presentation, and report
- Total marks: 300
  - Overall quality: 100 (group)
  - Presentation + Q&A: 100 (individually graded)
  - Report: 100 (group)
- Each group member should be present during final presentation and participate in the presentation
- Schedule of presentations will be shared later
- Guidelines for submitting code and report will be shared later

Dimensionality Reduction Techniques

Linear methods

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)

Non-linear methods

- t-Distributed Stochastic Neighbor Embedding (t-SNE)
- Uniform Manifold Approximation and Projection **(**UMAP)
- Multidimensional Scaling (MDS)
- ISOMAP
- Locally linear embedding (LLE)
- Laplacian Eigenmap (LE)

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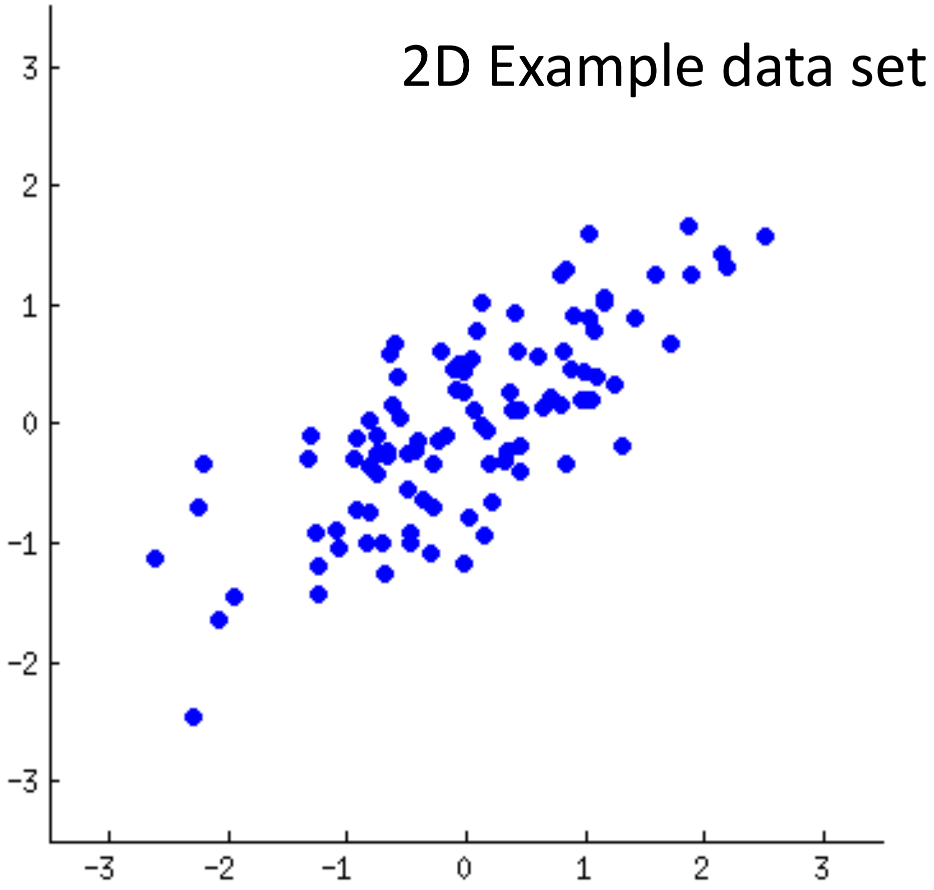
Principal Component Analysis (PCA)

- Unsupervised technique for extracting variance structure from high dimensional data to represent data in a lower dimensional space
  - An orthogonal (linear) projection of the data into a subspace so that the variance of the projected data is maximized
- <u>Useful for:</u>
  - Visualization
  - Further processing by machine learning algorithms
  - More efficient use of resources (e.g., time, memory, space)
  - In general: fewer dimensions → better generalization and modeling

Principal Component Analysis

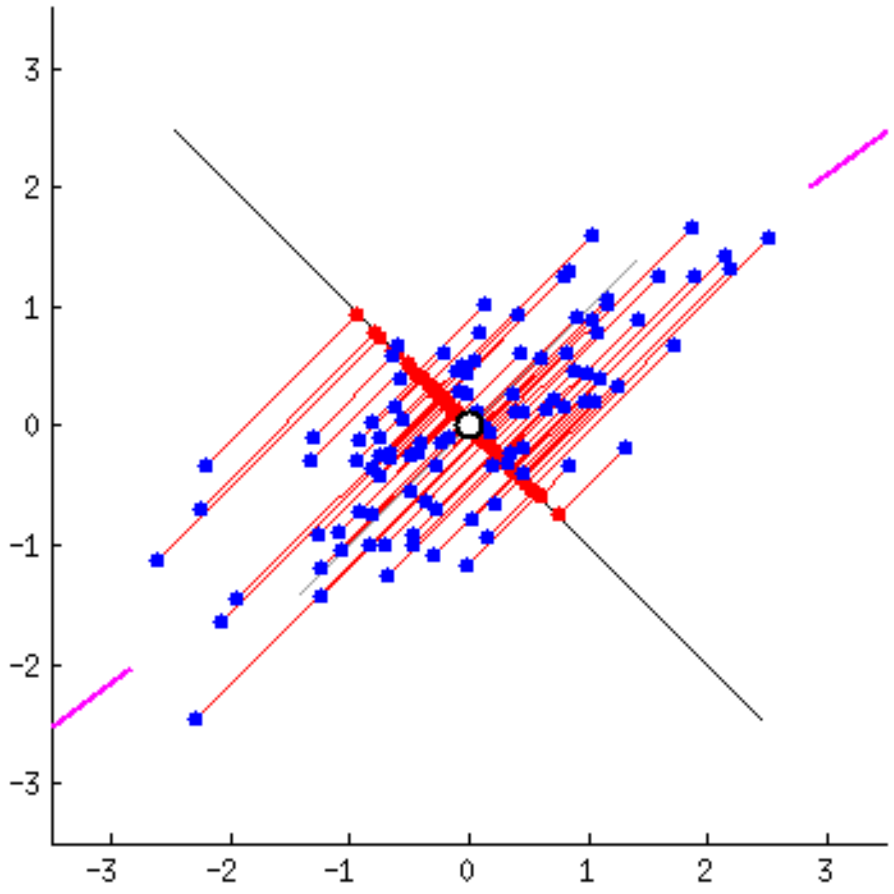
- A linear transformation that chooses a new coordinate system for the data such that greatest variance by any projection of the data comes to lie on the first axis (first principal component), the second greatest variance on the second axis, and so on….
- How to reduce data dimension with PCA?
  - Principal components are sorted in the order of their “explained variance” in the data
  - Eliminating later principal components allow dimension reduction

Principal Component Analysis



https://stats.stackexchange.com/questions/2691/making-sense-of-principal-component-analysis-eigenvectors-eigenvalues/140579

Principal Component Analysis



Animated view of how PCA works conceptually

Steps of PCA Algorithm

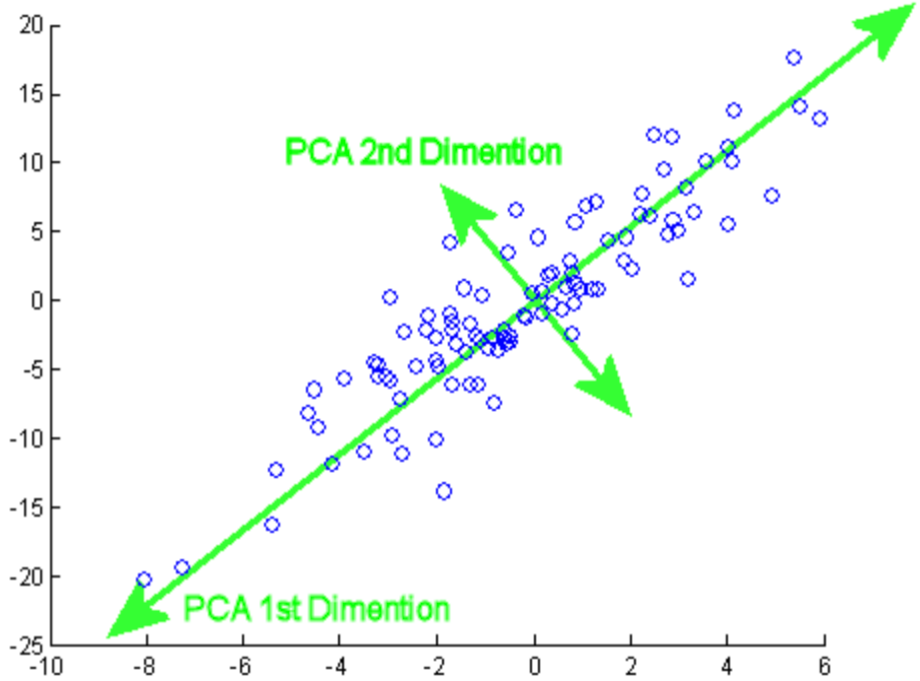
- Suppose we have a dataset with *n* records and *m* features for each record, i.e., data has *n* rows and *m* columns

1. Standardize the data: $x_{new} = \dfrac{x-\mu}{\sigma}$

2. Calculate the covariance matrix of the standardized data matrix
3. Calculate the eigen decomposition of the covariance matrix
   - Results in a list of eigenvalues and a list of eigenvectors
4. Sort eigenvalues in descending order to get a ranking for the eigenvectors (principal components) or axes of the new subspace

How to Project Data into a Lower Dimension?

- A total of k components (k < m) can be selected to create a projection subspace.
- The k eigenvectors are called principal components, that have the k largest eigenvalues
- Data can be projected into the k-dimensional subspace via matrix multiplication



PCA: Explained Variance

- Explained variance is a statistical measure of how much variance in a dataset can be attributed to each of the principal components
  - How much of the total variance is "explained" by each component
- Ordered (large to small) eigenvalues can help
- Explained variance for *i*th principal component:

$$\frac{\lambda_i}{\lambda_1 + \lambda_2 + \ldots + \lambda_n}$$

λ_i : ith eigen value of covariance matrix

Visualization of Data using PCA

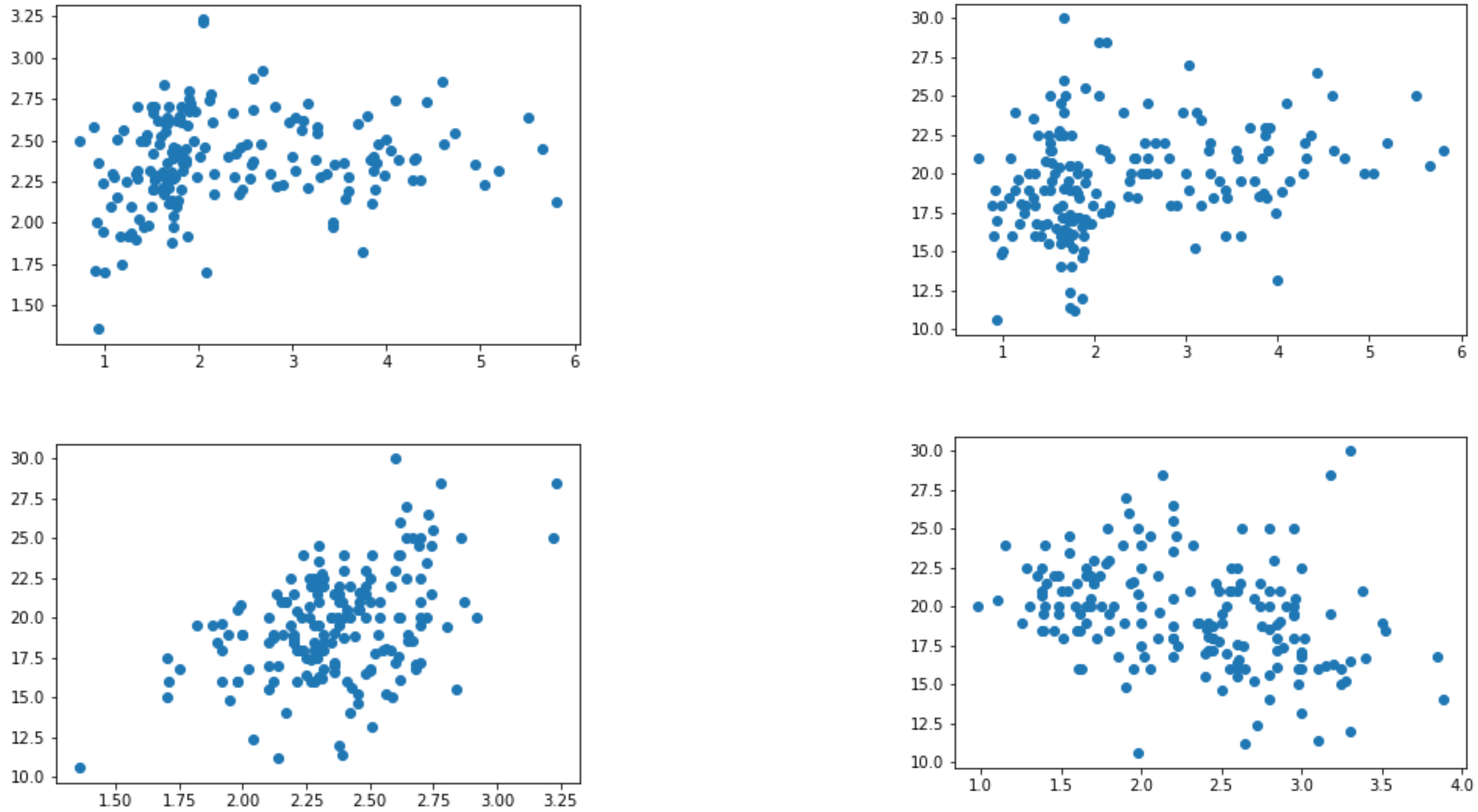
- Wine data set
  - 178 records, 13 features for each record

```
[[1.423e+01 1.710e+00 2.430e+00 ... 1.040e+00 3.920e+00 1.065e+03]
 [1.320e+01 1.780e+00 2.140e+00 ... 1.050e+00 3.400e+00 1.050e+03]
 [1.316e+01 2.360e+00 2.670e+00 ... 1.030e+00 3.170e+00 1.185e+03]
 ...
 [1.327e+01 4.280e+00 2.260e+00 ... 5.900e-01 1.560e+00 8.350e+02]
 [1.317e+01 2.590e+00 2.370e+00 ... 6.000e-01 1.620e+00 8.400e+02]
 [1.413e+01 4.100e+00 2.740e+00 ... 6.100e-01 1.600e+00 5.600e+02]]
```

https://archive.ics.uci.edu/ml/datasets/wine

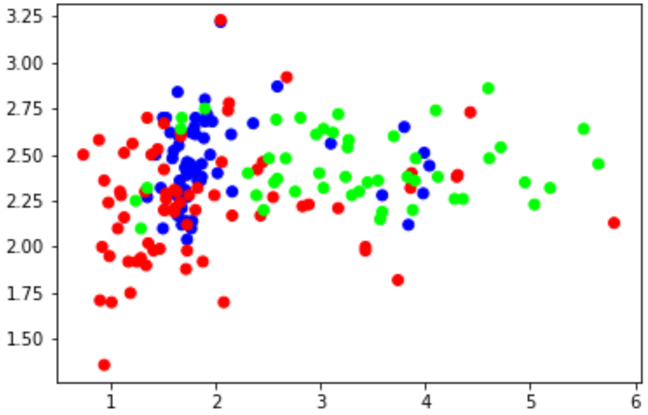
Visualization of Data using PCA

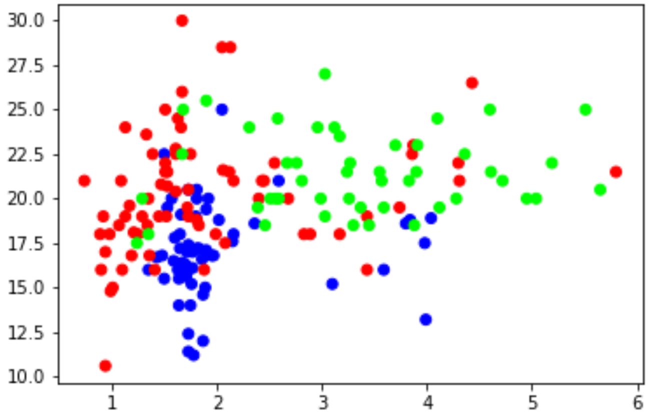
- Scatter plot for selected pairwise features (attributes)

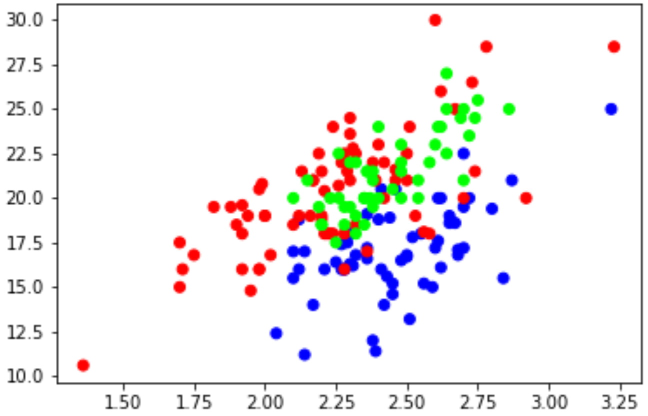


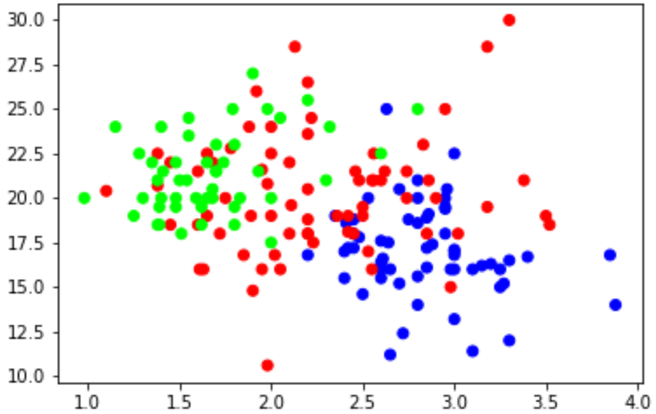
Visualization of Data using PCA

- Scatter plot for selected pairwise features (attributes)
  - Color points by class label





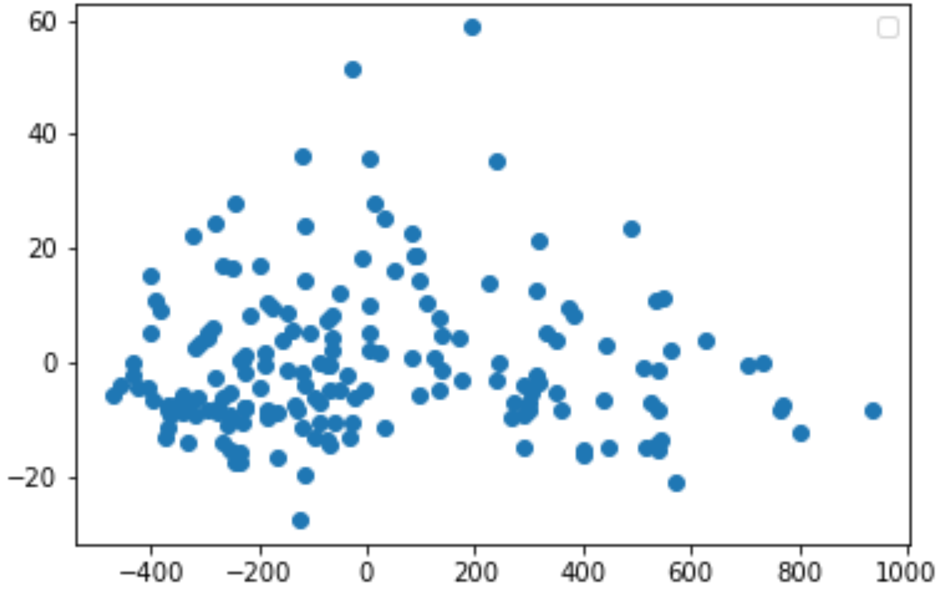




Visualization of Data using PCA

- Scatter plot using two first principal components as axes after PCA is applied

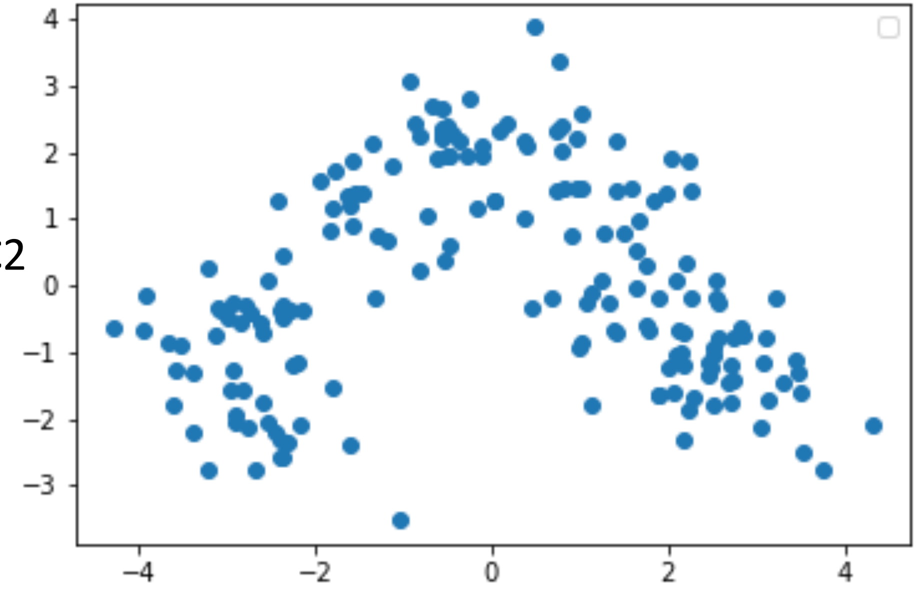
PC2



PC1

PCA with no standardization

PC2

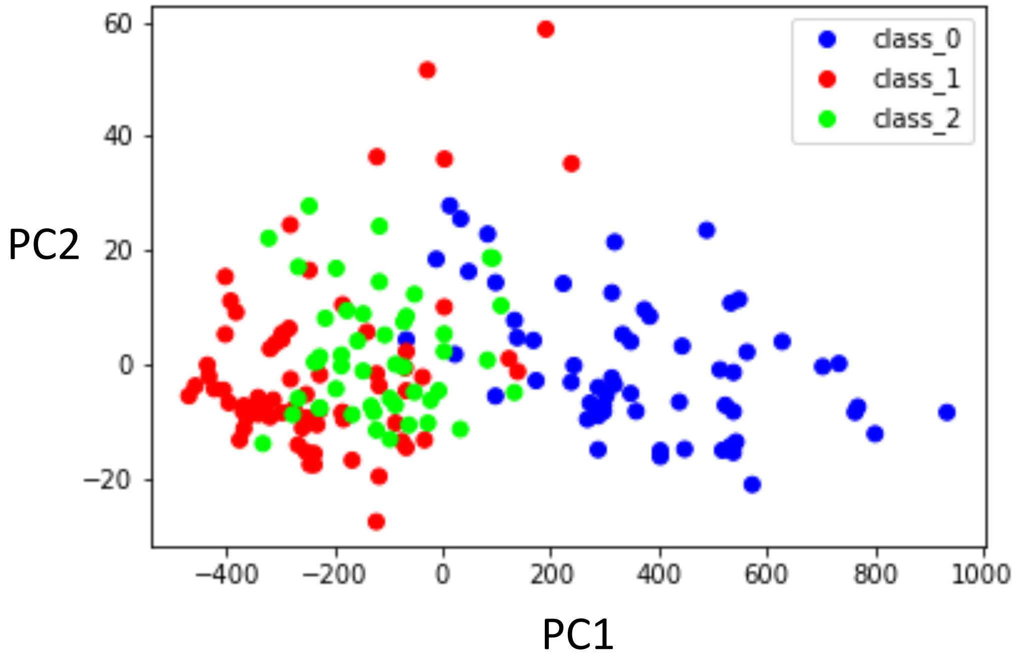


PC1

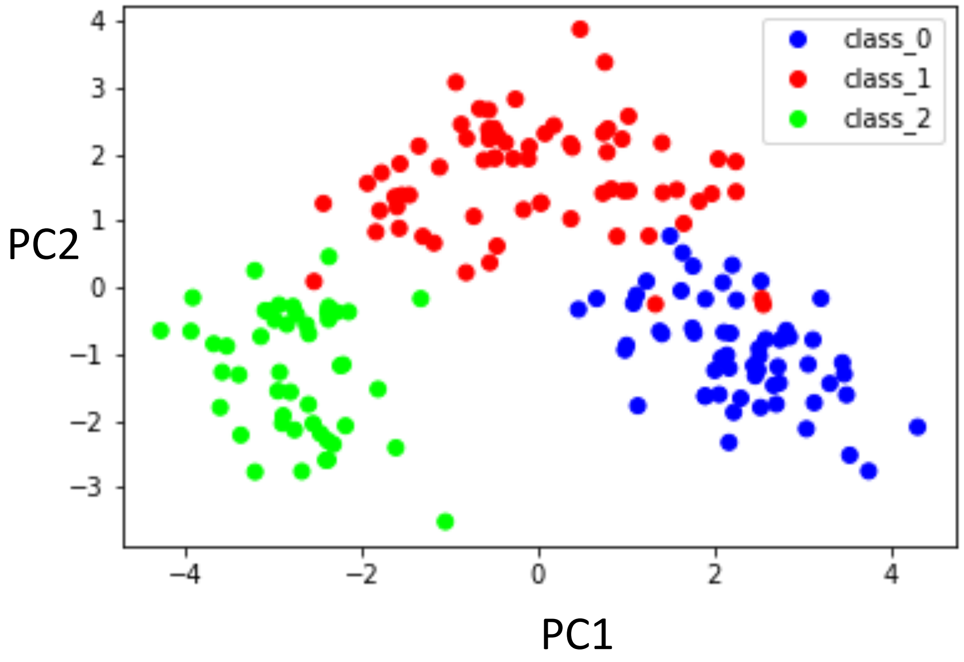
PCA with standardization

Visualization of Data using PCA

- Scatter plot using two first principal components as axes after PCA is applied
  - Color points by class label



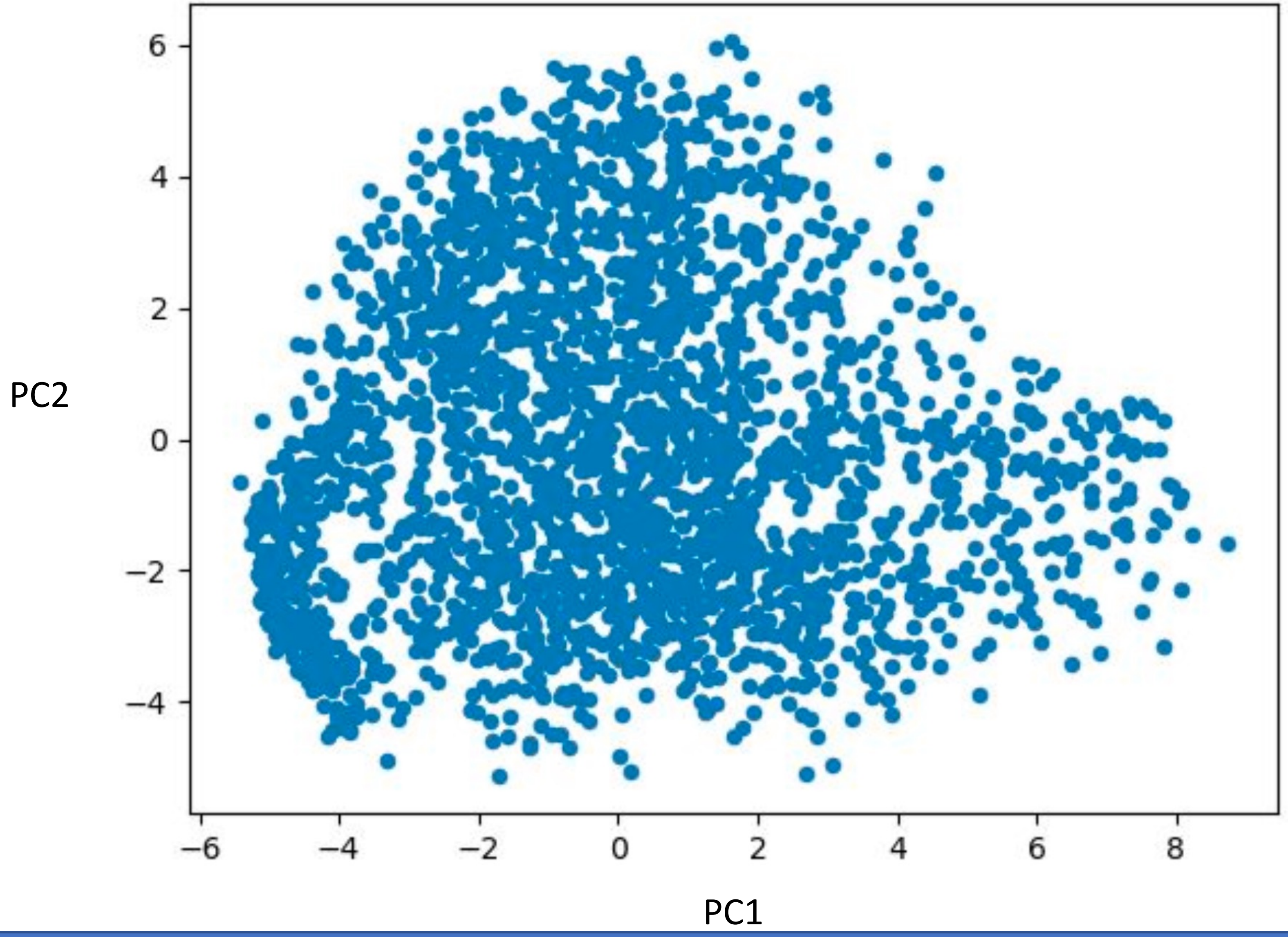
PCA with no standardization



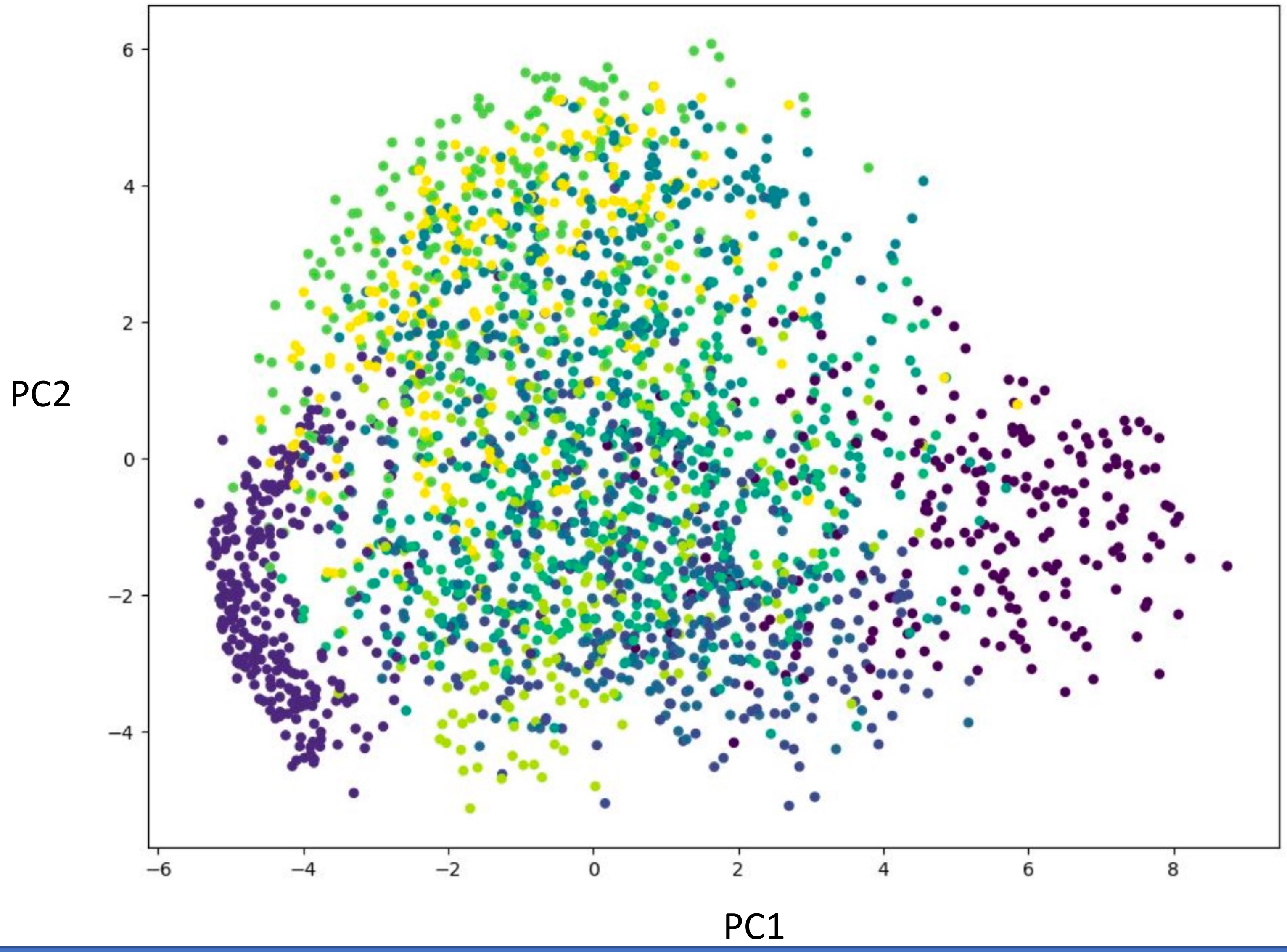
PCA with standardization

PCA on MNIST data

PCA on MNIST data



PCA on MNIST data



PCA is not able to separate classes clearly